



01 Executive Summary

SwarmOps solves the most critical bottleneck in drone fleet operations: real-time operational decision-making - which drone flies which mission, in what order and along which route, while dynamically managing battery limits, range and priorities. The technology is built on advanced optimization algorithms (VRPTW) and a distributed system that re-plans within seconds when conditions change in the field. The prototype is working, deployed on AWS and available for a live demonstration.

02 The Challenge

- Operational complexity and disconnected tools:** manufacturer (OEM) software flies a single aircraft superbly, but the tools do not talk to each other. Nothing provides synchronization and one unified operational picture at fleet level - who is doing what, what the battery state is, and whether the aircraft can complete the mission safely.
- The scale problem:** managing one drone is simple, but past a fleet of dozens the coordination problem becomes computationally intractable for a human operator. Analyzing battery, weight, time-window and airspace constraints in real time demands autonomous algorithms - without them the manual process collapses.
- The cost in the field:** critical flight time wasted, duplicate coverage, airframe wear, and worst of all - inaccuracy and confusion in the real-time operational picture.

03 The Solution & Product

VENDOR-NEUTRAL LAYER

A vendor-neutral coordination and optimization layer

- Assignment and routing algorithms (VRPTW):** optimal route computation, automatic mission allocation and real-time integration of charging / battery-swap stops - for maximum air time and minimum risk.
- Dynamic real-time re-planning:** continuous processing of telemetry and field events updates the mission board within seconds - on a technical fault, a battery draining faster than planned, loss of signal, or an urgent change in operational priorities.
- Unified control room and a coherent operational picture:** centralized fleet control, live observation feed to commanders and managers, and dynamic dashboards showing mission status, aircraft readiness and remaining air time in real time.
- Hardware agnostic:** seamless connection to mixed fleets from any manufacturer or protocol (such as MAVLink) - full use of the aircraft an organization already owns.
- Flexible, deployment-ready architecture:** a distributed system supporting rapid deployment in the cloud (AWS and equivalents), on local servers in closed networks (on-premise), or at edge points in the field - container-based (Docker & Kubernetes) with automated deployment pipelines (CI/CD) and full observability.

06 Team

Tony Verin (team lead) | Guy | Harel - three engineers, veterans of field and combat units, **two of them drone operators**, who designed, built and operate the entire system end to end: algorithms, cloud architecture, user interfaces and infrastructure.

Why We Built This

This is not a theoretical scenario for us. On 7 October 2023, during the fighting in the kibbutzim, two teams from the same company launched drones at the same time. With no coordination between them, both swept the same sector - one team identified hostiles, the other reported a friendly force in the same place. The contradiction in the situational picture delayed a critical operational decision. **That is the problem SwarmOps was built to solve.**

04 Applications & Market Potential

DUAL-USE

Defense, emergency & C4I	Unified operational picture, coordinated sweeps and rapid response, alongside intelligent fleet logistics (battery and airframe health), and an ML component for lessons learned, pattern recognition and sector-specific optimization.
Construction & infrastructure (ConTech)	Autonomous periodic mapping against project milestones - earthwork volume calculations, progress verification for releasing funds from banks and financiers, comparison of as-built conditions against engineering plans (BIM), and legal documentation to prevent overruns.
Infrastructure & energy	Periodic monitoring, defect scanning and preventive maintenance of power lines, oil and gas pipelines, national assets and solar installations.
Precision agriculture	Coordination of multispectral aerial surveys for pest detection, and scheduled autonomous spraying at scale.
Logistics & delivery	Autonomous site-to-site transport management and last-mile optimization across the supply chain.

05 Roadmap - 12 Months

Months 0-3

Connecting real aircraft (MAVLink), initial field trials and engagement with the Civil Aviation Authority.

Months 3-6

First pilot with a strategic partner / design customer to prove success metrics.

Months 6-12

Transition to a Multi-Tenant product, commercial pricing and a Pre-Seed / Seed round.

07 The Ask

As we move from prototype (PoC) to an operational/commercial product, we are looking for:

- A place in a **technology incubator / accelerator**, with business mentorship.
- Strategic introductions:** institutional operators, integrators and infrastructure companies.
- Pre-Seed support:** to acquire a drone fleet for live testing, cloud resources and regulatory/legal counsel.